

End-to-End Sales Forecasting & Demand Intelligence System

Executive Business Report

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Executive Summary

This project presents a complete Sales Forecasting and Demand Intelligence System developed using Python, Pandas, Statsmodels, Prophet, XGBoost, Scikit-learn, and Streamlit. The solution transforms historical retail sales data into business insights that support inventory planning, demand forecasting, and strategic decision-making. The system analyzes historical trends, identifies seasonal buying behavior, predicts sales for the next three months, detects unusual sales patterns, segments products according to demand, and provides an interactive dashboard for business users. The objective is to reduce inventory costs, improve product availability, and help management make proactive decisions using data rather than intuition.

Key Findings from Exploratory Data Analysis

- Technology generated the highest overall revenue, demonstrating consistently strong customer demand.
- Office Supplies showed stable purchasing behavior throughout the year, making it suitable for predictable inventory planning.
- Furniture experienced comparatively larger fluctuations in sales, indicating greater demand uncertainty.
- The West region recorded the highest sales performance, while the East region maintained stable year-over-year growth.
- Average shipping time remained consistent across regions, suggesting efficient logistics operations.
- Monthly analysis revealed strong year-end sales peaks, confirming the presence of seasonal demand.

Three-Month Sales Forecast

Three forecasting approaches—SARIMA, Prophet, and XGBoost—were developed and evaluated using MAE, RMSE, and MAPE. The selected model demonstrated the best balance between prediction accuracy and stability. Forecasts indicate continued healthy demand over the next three months with confidence intervals that provide a realistic range for expected sales. These predictions can be used to optimize purchasing, warehouse allocation, staffing, and replenishment schedules.

Top Anomalies Detected

1. A significant sales spike likely associated with festive promotions or seasonal discounts.
 2. A temporary decline in sales that may indicate stock shortages, logistics delays, or reduced customer activity.
 3. A short-lived increase in demand possibly caused by marketing campaigns or regional promotional events.
- Both Isolation Forest and Z-Score techniques identified similar unusual periods, increasing confidence in the anomaly detection results.

Product Demand Segmentation

Products were grouped into four demand clusters using K-Means clustering.

- High Volume – Maintain high inventory and prioritize replenishment.
- Growing Demand – Gradually increase stock levels to support future growth.
- Stable Demand – Continue existing inventory strategy with regular monitoring.
- Low Volume – Reduce excess inventory and consider targeted promotional activities.

Business Recommendations

1. Increase inventory before seasonal demand peaks to reduce stock-out risk.
2. Adopt forecast-driven inventory planning instead of relying only on historical averages.
3. Integrate anomaly monitoring into daily operations to quickly identify unexpected demand changes and respond proactively.

Business Value

The proposed system enables better demand planning, improves inventory utilization, reduces holding costs, minimizes lost sales due to stock shortages, and supports faster managerial decision-making through an interactive dashboard. The dashboard allows executives to monitor sales trends, compare forecasts, review anomalies, and evaluate product demand segments without requiring technical knowledge.

Risk and Limitation

Forecast accuracy depends on historical data and assumes that future market conditions remain reasonably similar. Unexpected events such as economic changes, supply-chain disruptions, competitor actions, or policy changes may reduce prediction accuracy. Regular model retraining with recent data is recommended to maintain reliable forecasts.

Conclusion

The End-to-End Sales Forecasting & Demand Intelligence System demonstrates how statistical forecasting, machine learning, anomaly detection, clustering, and business visualization can be combined into a practical decision-support solution. The project provides actionable insights that help retail organizations improve operational efficiency, optimize inventory management, and make informed strategic decisions.